



CAUSAL REPRESENTATION LEARNING

*Heterogeneous Treatment Effects
from Observational Data*

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ABSTRACT

Estimating heterogeneous treatment effects from observational data requires both causal identification assumptions and models that generalize across imbalanced treatment populations. This survey introduces the potential-outcomes framework, distinguishes identification from estimation, and reviews classical adjustment through outcome regression, propensity scores, weighting, and doubly robust methods. It then focuses on causal representation learning. TARNet shares predictive structure between treatment groups, while Counterfactual Regression adds an Integral Probability Metric penalty motivated by a generalization bound; Maximum Mean Discrepancy and Wasserstein distance are presented as its principal instantiations. BCAUSS and HERMES illustrate subsequent directions based on self-supervised balancing and effect-aware contrastive geometry. Finally, the survey examines large-language-model embeddings as semantic side information. Rather than treating generated outputs as causal estimates, it argues for using semantic representations as priors or regularizers within procedures whose identification assumptions and statistical guarantees remain explicit. The resulting perspective separates what learned representations can improve counterfactual generalization and inductive bias from what they cannot supply: unmeasured-confounding control, structural overlap, or counterfactual ground truth.

Keywords: causal inference; heterogeneous treatment effects; counterfactual estimation; representation learning; observational data; large language models

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1 Introduction

Scientific and operational processes require decisional procedures that often need more than predicting what is likely to happen. A physician choosing between therapies, for example, is interested in how the same patient’s outcome would change under each available treatment. This is a *causal* question because it compares alternative interventions, including outcomes that cannot all be observed. A model may accurately predict the outcome received under the actual treatment while remaining unreliable about the outcome that would have followed from a different decision.

The Potential-Outcomes Framework formalizes this problem by associating each unit with both an outcome under treatment and under control group (Rubin 2005; Holland 1986). The difference between the predicted values defines a causal effect. Here lies the main problem: only one potential outcome is observed for each unit.

Randomized experiments address this missing-counterfactual issue relying on randomization of patients assignment in both groups to make them comparable on the **Average Treatment Effect** (ATE). In observational studies, by contrast, treatment assignment may depend on patient characteristics, institutional policies, or other pre-treatment variables. The treated and control samples can therefore differ substantially, and an observed association may combine the treatment effect with pre-existing differences between the groups.

This challenge becomes more troublesome when treatment effects are heterogeneous. Here, an average effect summarizes the response of a population, but it may conceal subgroups that benefit, are unaffected, or are harmed. The **Conditional Average Treatment Effect** (CATE) describes how the expected causal effect varies with observed covariates. Estimating this function requires learning two treatment-specific response surfaces, even though each observation supplies a factual outcome for only one of them. Under covariate imbalance, each surface must also be evaluated on units drawn from a distribution that may differ from the one on which it was trained. Counterfactual estimation can therefore be viewed, after the causal assumptions have been stated, as a problem of generalization under distribution shift.

Causal Representation Learning addresses this statistical difficulty by transforming high-dimensional covariates into a latent space designed for counterfactual prediction. TARNet (Shalit et al. 2017) shares predictive structure between groups through a common representation while providing treatment-specific outcome heads. Counterfactual Regression (CFR) (Shalit et al. 2017) additionally penalizes discrepancies between treated and control representations, connecting the resulting objective to a generalization bound based on Integral Probability Metrics (Müller 1997). Maximum Mean Discrepancy and Wasserstein distance provide two ways to quantify this discrepancy. The central trade-off is that the representation should make the groups sufficiently comparable without discarding variables that predict outcomes or modify the treatment effect.

Later methods enrich the signal used to shape this latent space. BCAUSS (Tesei et al. 2023) directly optimizes a self-supervised covariate-balancing criterion, while HERMES (Zaccagnino et al. 2026) uses contrastive pairs based on estimated treatment-response similarity. These approaches shift attention from global distribution matching toward the geometry of local relationships among units. They also raise a broader question: whether information not contained in the observed numerical covariates can provide a useful inductive bias.

Large language models (LLMs) offer one possible source of such information. Recent literature considers both using LLMs in causal tasks and applying causal methods to understand or improve language models (Liu et al. 2025).

Directly generated causal judgments or counterfactuals, however, may not by themselves supply the identification assumptions or statistical validation required for treatment-effect estimation (Nitish Joshi and He 2024). A more conservative possibility is to use LLM-derived embeddings as semantic side information. Work in genomics, for instance, has used embeddings of textual gene descriptions to define informative subspaces for hypothesis tests while preserving frequentist error control (Bryan et al. 2025). This example does not establish a method for CATE estimation, but it illustrates how semantic information may guide a statistical procedure without replacing its formal guarantees.

This survey develops these arguments in stages:

- Section 2 introduces potential outcomes, causal estimands, and the assumptions required for identification.
- Section 3 reviews outcome regression, propensity-score adjustment, weighting, and doubly robust estimation, emphasizing their common dependence on overlap between treatment groups.
- Section 4 presents TARNet and CFR, together with the IPM-based generalization argument and its MMD and Wasserstein instantiations.
- Section 5 discusses BCAUSS and contrastive effect-aware representations.
- Section 6 then examines semantic embeddings and LLM-derived priors as possible auxiliary inputs.
- Sections 7 and 8 summarize the methodological trade-offs and the limits that representation learning cannot remove.

2 Potential Outcomes and Causal Identification

Causal inference aims to quantify how an outcome would change under alternative interventions. This goal differs from ordinary prediction: a predictive model describes the outcome that is likely to be observed, whereas a causal analysis compares outcomes associated with decisions that may not have actually been taken. The potential-outcomes framework provides a formal language for defining such comparisons and for stating the assumptions under which they can be recovered from observed data (Rubin 2005; Holland 1986).

Throughout this work, we consider a static binary treatment. For each unit i , let $X_i \in \mathcal{X}$ denote a vector of pre-treatment covariates, $T_i \in \{0, 1\}$ the assigned treatment, and Y_i the observed outcome, with $Y_i = 1$ for a success, 0 for a failure. The treatment value $T_i = 1$ represents the intervention of interest, whereas $T_i = 0$ represents the control condition. Although this terminology originates from clinical trials, the treatment may more generally represent any actionable exposure, policy, or decision whose causal effect is under study. Likewise, the term patient is used as synonym for an individual or a unit which receives one of the treatments.

2.1 Potential Outcomes and the Missing Counterfactual

For each unit i , the framework defines two potential outcomes:

$$Y_i(1), \quad Y_i(0). \tag{1}$$

$Y_i(1)$ is the outcome that would be observed if the unit received treatment and $Y_i(0)$ is the outcome that would be observed under control. Both quantities are conceptually defined

before the treatment is observed, regardless of which treatment the unit actually receives.

Only the potential outcome corresponding to the assigned treatment is observed. Under consistency, the observed outcome can be written as:

$$Y_i = T_i Y_i(1) + (1 - T_i) Y_i(0). \quad (2)$$

If $T_i = 1$, Equation (2) reduces to $Y_i = Y_i(1)$, while $Y_i(0)$ remains unobserved. On the other hand, if $T_i = 0$, the observed outcome is $Y_i(0)$, while $Y_i(1)$ is missing. The unobserved potential outcome is commonly referred to as the *counterfactual outcome*.

The impossibility of observing both potential outcomes for the same unit under the same conditions is known as the *fundamental problem of causal inference* (Holland 1986).

2.2 Randomized Controlled Trials and Observational Studies

Researchers have developed several study designs and statistical methods to address the challenge posed by the fundamental problem of causal inference.

The randomized controlled trial (RCT) is widely regarded as the gold standard for evaluating the causal effects of interventions (Hariton and Locascio 2018). In an RCT, treatment is assigned at random, making treatment assignment independent of the potential outcomes by design. Differences in average outcomes between the treatment and control groups can therefore be attributed to the intervention, up to sampling variability and subject to conditions such as consistency and no interference. Nevertheless, even in a perfectly randomized experiment, randomization does not reveal both potential outcomes for any one patient. Instead, it makes the treatment groups comparable in expectation, allowing average potential outcomes to be estimated through between-group comparisons. Despite their strengths, RCTs can be costly and time-consuming, and in some settings they may be infeasible or unethical (Fernainy et al. 2024). For example, researchers cannot deliberately assign participants to harmful exposures, and randomization may be unethical when it would withhold an established effective treatment from the control group. In such settings, researchers often rely on observational studies, in which treatment assignment is not controlled by the researcher and may depend on patient characteristics related to the outcome. Consequently, observed differences between treatment groups may reflect both the causal effect of treatment and pre-existing differences between the groups. Recovering causal effects from observational data therefore requires additional assumptions and adjustment for confounding.

These distinctions are central to the remainder of this work. The problem is not merely one of predicting an ordinarily missing value. Counterfactual outcomes are missing according to the treatment-assignment mechanism, and their estimation requires assumptions connecting the observed treatment groups to the hypothetical interventions of interest.

2.3 Causal Estimands

A causal estimand is the population quantity that an analysis seeks to identify and estimate. Different estimands answer different scientific questions and should not be treated as interchangeable.

At the unit level, the causal effect of treatment for unit i is:

$$\tau_i = Y_i(1) - Y_i(0). \quad (3)$$

This contrast is conceptually meaningful, but, as previously stated, it depends on two outcomes that cannot be jointly observed. Standard randomized or observational data also do not identify the joint distribution of $Y(0)$ and $Y(1)$. Consequently, the unit-level effect is not generally identifiable without additional structural assumptions beyond those used to identify conditional mean effects.

Averaging over the target population gives the ATE:

$$\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)]. \quad (4)$$

ATE compares two hypothetical interventions on the same target population: assigning every unit to treatment and assigning every unit to control. It may nevertheless conceal substantial heterogeneity: a small average effect can be the result of benefits in one subgroup, negligible effects in another, and harm in a third. More importantly, it carries little to no information to the outcome of individual patients.

Estimating how treatment effects vary with observed characteristics is commonly referred to as heterogeneous treatment-effect estimation (HTE) (Angrist 2003). Rather than summarizing the intervention through a single population average, this task aims to recover the function τx , thereby identifying covariate-defined groups that may respond differently to the same treatment.

Treatment-effect heterogeneity is described through the CATE:

$$\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]. \quad (5)$$

CATE is an expected treatment effect among patients sharing the covariate profile $X = x$. It is therefore a function of observed pre-treatment characteristics rather than the unobservable effect of a particular unit.

Some representation-learning papers use the term *Individual Treatment Effect* (ITE) for a treatment-effect function evaluated at an individual's covariates. Here, we will denote *CATE* as the conditional expectation in Equation (5), while *unit-level effect* will denote the generally unobservable contrast in Equation (3).

Given this notation, the CATE can also be written through the two conditional response functions:

$$\mu_t(x) = \mathbb{E}[Y(t) \mid X = x], \quad t \in \{0, 1\}. \quad (6)$$

The $\tau(x)$ value is then:

$$\tau(x) = \mu_1(x) - \mu_0(x). \quad (7)$$

2.4 CATE Assumptions

The term *identification* concerns whether a causal estimand can be expressed as a functional of the observed-data distribution. It is logically distinct from estimation: identification states what could be recovered with unlimited data from the same data-generating process, whereas estimation concerns how accurately that quantity can be learned from a finite sample. The following assumptions are required to identify the CATE from observational data.

2.4.1 Consistency

Consistency connects potential and observed outcomes. If a unit receives treatment t , its observed outcome must equal the corresponding potential outcome:

$$T_i = t \implies Y_i = Y_i(t). \quad (8)$$

This condition requires the intervention $T = t$ to be sufficiently well defined.

2.4.2 No Interference and SUTVA

The no-interference assumption states that the potential outcome of one unit does not depend on treatments assigned to other units. If $\mathbf{T}_{-i}$ denotes the assignments of all units except i , then

$$Y_i(t_i | \mathbf{T}_{-i}) = Y_i(t_i). \quad (9)$$

Consistency and no interference are commonly grouped under the Stable Unit Treatment Value Assumption, or SUTVA.

2.4.3 Conditional Exchangeability

In observational studies, treatment assignment may depend on characteristics that also influence the outcome. Conditional exchangeability assumes that all such common causes have been adequately represented by the observed pre-treatment covariates:

$$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X. \quad (10)$$

Within strata defined by X , treatment assignment then carries no additional information about the potential outcomes. The condition is also called *unconfoundedness*, *ignorability*, or *no unmeasured confounding* (Rosenbaum and Rubin 1983; Imbens and Wooldridge 2009). Conditional exchangeability concerns potential outcomes, not the observed outcome. Also, unlike randomization, this assumption cannot generally be established from observed data alone. Its plausibility depends on solid knowledge of the treatment-assignment process and on whether the available covariates contain the relevant pre-treatment common causes. Increasing model complexity cannot compensate for an omitted confounder.

2.4.4 Positivity

Positivity requires each treatment level to occur with non-zero probability for every covariate profile in the target population:

$$0 < \mathbb{P}(T = 1 \mid X = x) < 1 \quad \text{for all } x \text{ such that } p_X(x) > 0. \quad (11)$$

It is useful to distinguish structural and practical positivity problems. A structural violation occurs when one treatment is impossible for some covariate profiles, so the corresponding counterfactual cannot be identified without extrapolation or a change in target population. A practical problem occurs when both treatments are possible but one is extremely rare in a region. The estimand may remain formally identified, yet estimation can have high variance and depend strongly on modelling assumptions.

Representation learning may improve empirical comparability by mapping units into a lower-dimensional space. It cannot create information about interventions structurally absent from the data. Conditional exchangeability together with positivity is often called *strong ignorability* (Rosenbaum and Rubin 1983), although keeping the conditions separate clarifies their distinct roles.

2.5 Identification of Conditional Treatment Effects

The assumptions above connect the causal estimation to the observed-data distribution. For $t \in \{0, 1\}$, conditional exchangeability gives

$$\mathbb{E}[Y(t) \mid X = x] = \mathbb{E}[Y(t) \mid T = t, X = x] \quad (12)$$

Among units who receive treatment t , consistency allows the potential outcome to be replaced by the observed outcome:

$$\mathbb{E}[Y(t) \mid T = t, X = x] = \mathbb{E}[Y \mid T = t, X = x] \quad (13)$$

Positivity ensures that the conditional expectation on the right is defined for both treatment levels throughout the support of interest. Combining the steps yields

$$\mathbb{E}[Y(t) \mid X = x] = \mathbb{E}[Y \mid T = t, X = x] \quad (14)$$

Consequently, the CATE is identified as:

$$\begin{aligned} \tau(x) &= \mathbb{E}[Y(1) - Y(0) \mid X = x] \\ &= \mathbb{E}[Y \mid T = 1, X = x] - \mathbb{E}[Y \mid T = 0, X = x] \end{aligned} \quad (15)$$

Averaging over the target covariate distribution similarly identifies the ATE:

$$\tau_{\text{ATE}} = \mathbb{E}_X[\mathbb{E}[Y \mid T = 1, X] - \mathbb{E}[Y \mid T = 0, X]]. \quad (16)$$

These formulas do not prescribe a particular estimator. Outcome regression, matching, propensity-score weighting, doubly robust methods, and representation-learning models are different strategies for estimating the observed-data quantities above. Their causal interpretation still depends on the identification assumptions.

Identification also does not imply easy estimation. When treated and control units occupy different regions of the covariate space, one or both response functions must be learned using limited local evidence. This becomes particularly severe for high-dimensional covariates and HTEs.

2.6 Assumption Violation and Biases

2.6.1 Conditional Exchangeability Violation

Violations of conditional exchangeability often depends on the available data and trial type. Observational studies, in particular, suffer from two main biases due to this kind of violation, confounding bias and selection bias.

Confounding Bias Confounding occurs when a variable is a common cause of treatment and outcome. In the example discussed by Skelly et al. (2012), about an observational comparison of vertebroplasty and conservative care, patients receiving vertebroplasty had different smoking patterns or disease severity from those receiving conservative care. These factors also affected the risk of subsequent fractures, such that the crude treatment comparison combined the effect of vertebroplasty with the effect of those pre-existing differences. Vertebroplasty initially appeared to double the risk of subsequent fracture, but the association was close to null after stratifying by smoking status, illustrating how uncontrolled confounding can produce a misleading causal conclusion.

Confounding can sometimes be addressed by randomization or, in observational studies, by measuring and appropriately adjusting for sufficient pre-treatment confounders through stratification, regression, standardization, or inverse-probability weighting. However, adjustment cannot generally remove bias from unmeasured or poorly measured confounders. Consequently, if conditional exchangeability is not plausible after adjustment, the estimated effect may be overestimated, underestimated, or even reversed, weakening the study's internal validity.

Selection Bias Selection bias occurs when inclusion in the analyzed sample depends on variables related to treatment and outcome. A common example is the collider bias (Hernán 2017): if both treatment and another cause of the outcome influence study participation, conditioning on participation can create a noncausal association between treatment and outcome. The study may then produce a biased estimate even when no confounding was present in the original population.

Selection bias therefore affects validity by making the analyzed participants systematically different from the target population, or by inducing an artificial association within the selected sample. Appropriate study design, careful definition of the target population and retention of participants can help address the problem, provided the selection mechanism is sufficiently understood and measured.

2.6.2 SUTVA Violations

SUTVA violations are often linked to bad trial design therefore they may more easily be controlled and mitigated by improving the study enrollment and treatment definition.

Consistency Violation A consistency violation occurs when an exposure variable in an observational study is defined too broadly, encompassing multiple distinct versions of a treatment that actually produce different outcomes. An exposure lacking specificity implies a severe bottleneck in translation. Even if a strong statistical association is found, the inherent ambiguity makes it impossible to design a precise, effective intervention.

To illustrate, consider the example from Rehkopf et al. (2016) on how "education" is routinely used as an exposure in social epidemiology. Most research operationalizes education simply as total years of schooling completed or degrees obtained. However, this generic metric bundles together vastly different potential interventions, such as improving teacher quality, reducing class sizes, or increasing the specific focus on functional literacy. Treating "years of education" as a single, uniform exposure creates a consistency violation, rendering the findings practically inadequate for designing targeted, future-oriented public health policies.

No Interference Violation A violation of the no interference assumption occurs when one person's outcome depends not only on their own treatment, but also on the treatment received by others, such that the usual comparison between treated and untreated individuals may not represent a single causal effect.

Hudgens and Halloran (2008) illustrate this with a cholera-vaccine trial. In areas with low vaccine coverage, the estimated direct effect of the vaccine was much higher than in areas with high coverage. This was because vaccination reduced transmission and also protected unvaccinated people. Thus, ignoring interference could make the vaccine's direct effect appear smaller and would fail to capture its population-level benefit.

The paper also shows that violating no interference does not always make causal inference invalid. If the study is designed to account for interference and the causal question concerns treatment strategies, such as high versus low vaccination coverage, direct, and indirect effects can still be estimated. However, without assumptions describing the interference structure, conventional individual-level effects may be poorly defined and standard uncertainty estimates may be invalid.

The next section examines classical adjustment strategies and interprets the remaining challenges as counterfactual generalization under distribution shift.

3 Estimation from Observational Data

The identification formula in Equation (15) expresses the conditional treatment effect through two observed-data regression functions. It does not, however, imply that these functions can be estimated equally well throughout the covariate space. In observational studies, treatment assignment is determined by clinical decisions, individual preferences, institutional constraints, or other characteristics of the units under study. Treated and control samples may therefore represent systematically different populations, making counterfactual estimation substantially harder than ordinary supervised prediction.

3.1 Treatment Assignment and Covariate Imbalance

In a randomized experiment, treatment assignment is independent of pre-treatment characteristics by design. The conditional treatment probability which can be described as:

$$e(x) = \mathbb{P}(T = 1 \mid X = x). \quad (17)$$

In an observational study, instead, it may vary considerably across the covariate space. This probability is known as Propensity Score (PS) (Rosenbaum and Rubin 1983) and it is widely used in causal inference (see Section 3.3). Consequently, the treated and control covariate distributions generally differ:

$$p(X \mid T = 1) \neq p(X \mid T = 0). \quad (18)$$

A naive comparison of observed group means does not generally identify a causal effect. For example, the observed difference can be decomposed as

$$\begin{aligned} \mathbb{E}[Y \mid T = 1] - \mathbb{E}[Y \mid T = 0] &= \\ \mathbb{E}[Y(1) - Y(0) \mid T = 1] + \left(\mathbb{E}[Y(0) \mid T = 1] - \mathbb{E}[Y(0) \mid T = 0] \right). \end{aligned} \quad (19)$$

The first term is the ATE among the treated, whereas the second compares the untreated potential outcomes of two different populations. Under unconditional exchangeability the second term vanishes. In observational data it may instead reflect differences in baseline risk, severity, access to treatment, or other causes of treatment assignment. This is the confounding component of the naive association.

The conditional exchangeability assumption from Section 2.4 states that adjustment for the observed covariates is sufficient to remove this bias. The remaining task is to solve the purely statistical problem of estimating the response function without over-extrapolating.

3.2 Outcome Regression

Outcome-regression methods estimate the treatment-specific conditional means:

$$\mu_t(x) = \mathbb{E}[Y \mid T = t, X = x], \quad t \in \{0, 1\}. \quad (20)$$

This regression can be computed by any statistical estimator by producing the following difference:

$$\hat{\tau}_{\text{OR}}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x). \quad (21)$$

The two functions may be estimated with parametric regression, trees, ensembles (Athey et al. 2015; Athey and Imbens 2016), neural networks, or other supervised-learning algorithms (Hartford et al. 2016). A single model can include treatment as an input to differentiate the outcome, while a two-model strategy fits on the data of each treatment group separately. These constructions underlie several meta-learning approaches to HTE estimation.

The rationale for outcome regression is that it directly targets the quantities appearing in the CATE identification formula. Its main limitation, however, is that estimating the effect for each unit requires predictions from both response surfaces, whereas the observed outcome provides information about only one of them. If treated and control units occupy different regions of the covariate space, $\hat{\mu}_1(x)$ must be extrapolated to covariate profiles represented mainly among controls, while $\hat{\mu}_0(x)$ must be extrapolated to profiles represented mainly among treated units. Flexible prediction methods can reduce approximation error in well-supported regions, but they cannot compensate for a lack of counterfactual support. Consequently, misspecification or extrapolation error in either response surface may bias the estimated treatment effect (Imbens and Wooldridge 2009).

3.3 Propensity Score

A complementary strategy models the treatment-assignment mechanism rather than the outcome. The propensity score in Equation (17) is a balancing score: Rosenbaum and Rubin (1983) showed that

$$T \perp\!\!\!\perp X \mid e(X). \quad (22)$$

Thus, among patients with the same PS, the distribution of observed covariates is the same in the treated and control groups. In other words, when conditioning covariates on the PS, it is not possible to discriminate the belonging of a patient to a specific group. Moreover, if treatment is conditionally exchangeable given X , this also holds given the true PS. This result reduces a potentially high-dimensional adjustment problem to adjustment for a scalar summary of the treatment-assignment mechanism.

The PS can be used for matching, stratification, covariate adjustment, or weighting.

3.4 Inverse Probability Weighting

Inverse Probability Weighting (IPW) uses the PS to reweight the observed sample according to the inverse probability of each unit receiving its actual treatment. Patients whose observed treatment was unlikely given their covariates receive larger weights, whereas units whose treatment was highly probable receive smaller weights.

Under correct assumptions and PS, this reweighting constructs a pseudo-population in which the measured covariate distribution is balanced across treatment groups. Comparisons of weighted outcomes can then be used to estimate population-level causal effects.

First, it is assigned to each observation a weight, such that:

$$w_i = \frac{T_i}{e(X_i)} + \frac{1 - T_i}{1 - e(X_i)}. \quad (23)$$

The weighted sample represents a pseudo-population in which measured covariates are balanced across treatment groups. A basic IPW-based estimator of the ATE is:

$$\hat{\tau}_{\text{IPW}} = \frac{1}{n} \sum_{i=1}^n \left[\frac{T_i Y_i}{\hat{e}(X_i)} - \frac{(1 - T_i) Y_i}{1 - \hat{e}(X_i)} \right]. \quad (24)$$

Weighting makes the connection between positivity and finite-sample stability explicit. If $e(X_i)$ is close to zero for a treated unit, or close to one for a control unit, the corresponding weight becomes large. A small number of observations can then dominate the estimate, increasing variance and sensitivity to propensity-model misspecification. Weight truncation, stabilized weights, overlap restrictions, and balance diagnostics are often used in practice, but they change the bias-variance trade-off and may implicitly modify the target population.

The PS is therefore a powerful compression of treatment-assignment information, not a complete solution to HTE estimation. It describes how treatment was assigned but does not model the outcome surface, and a scalar score may not preserve the geometry needed to estimate complex effect modification.

3.5 Doubly Robust Estimation

Outcome regression and propensity weighting fail in complementary ways. Doubly robust estimators (Funk et al. 2011) combine estimates of both functions so that consistency can be retained when either the outcome model or the propensity model is correctly specified, provided that the remaining regularity and identification conditions hold (Bang and Robins 2005; Imbens and Wooldridge 2009).

A widely used doubly robust method is *Augmented Inverse Probability Weighting* (AIPW) (Glynn and Quinn 2010). For each unit, it is defined the AIPW score as:

$$\hat{\psi}_i = \hat{\mu}_1(X_i) - \hat{\mu}_0(X_i) + \frac{T_i(Y_i - \hat{\mu}_1(X_i))}{\hat{e}(X_i)} - \frac{(1 - T_i)(Y_i - \hat{\mu}_0(X_i))}{1 - \hat{e}(X_i)}, \quad (25)$$

The first term in Equation (25),

$$\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i),$$

is the treatment-effect estimate obtained from outcome regression. The remaining terms correct this contrast using the residual between the observed outcome and the prediction corresponding to the treatment actually received. These residuals are weighted by the inverse estimated probability of receiving that treatment, so observations from poorly represented treatment assignments receive greater influence. Double robustness does not mean that arbitrary misspecification of both components is harmless, nor does it remove instability caused by severe lack of overlap.

The ATE is then estimated by averaging these unit-level scores:

$$\hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i. \quad (26)$$

Related orthogonal and pseudo-outcome methods extend this principle to conditional treatment-effect estimation. They construct regression targets that are locally insensitive, to first order, to errors in the estimated nuisance functions. These methods provide important alternatives to representation learning, but a comprehensive treatment of semiparametric CATE estimation lies beyond the scope of this survey.

3.6 Counterfactual Prediction as Distribution Shift

The preceding methods reveal a common difficulty. To estimate $\mu_t(x)$, outcomes under treatment t are observed only for units drawn from their respective distribution:

$$p_t(x) = p(X = x \mid T = t). \quad (27)$$

The fitted model must then be evaluated for units from both treatment groups. In particular, predicting the missing treated outcome of a control unit applies a model learned from $p_1(x)$ to covariate values drawn from $p_0(x)$, while predicting the missing control outcome of a treated unit reverses these roles. Whenever Equation (18) holds, counterfactual prediction therefore involves a distribution shift between the population supplying factual training outcomes and the population on which those predictions are required.

This perspective does not replace the causal identification assumptions. Conditional exchangeability is still needed to interpret the treatment-specific response surfaces causally, and positivity is still needed to ensure that relevant comparisons are supported by the data. It does, however, expose an additional learning problem: even when the effect is identified, finite-sample counterfactual predictions may generalize poorly because the treatment groups occupy different regions of a high-dimensional input space.

Causal representation learning addresses this problem by learning a transformation $\Phi(X)$ that supports accurate factual outcome prediction while reducing the discrepancy between the treated and control distributions in the representation space.

The next section develops this idea through TARNet and Counterfactual Regression, and explains how distributional distances enter a generalization bound for HTE estimation (Shalit et al. 2017).

4 Causal Representation Learning

The previous section interpreted counterfactual estimation as a prediction problem under treatment-dependent covariate shift. Causal representation learning addresses the problem by replacing the original covariates with a learned representation in which two objectives are pursued jointly: the representation should retain enough information to predict factual outcomes, while making treated and control units more comparable. The framework developed by Shalit et al. (2017) works on this intuition by connecting HTE estimation error to factual prediction error and to a discrepancy between the treatment-group distributions.

This section first introduces the shared-representation architecture underlying TARNet. It then defines the factual and counterfactual risks used in the generalization analysis, presents Integral Probability Metrics and their MMD and Wasserstein instances, and finally derives the CFR objective from the resulting upper bound.

4.1 Learning Treatment-Specific Response Functions

Representation-learning methods replace the original covariates X with a learned representation $\Phi(X)$. This representation is shared across treatment groups and is then passed to two treatment-specific outcome models, h_0 and h_1 . The two models estimate the outcome under control and treatment, respectively:

$$\hat{\mu}_0(x) = h_0(\Phi(x)), \quad \hat{\mu}_1(x) = h_1(\Phi(x)). \quad (28)$$

The conditional treatment effect is obtained by comparing these two predictions:

$$\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x) \quad (29)$$

The role of Φ is therefore not to estimate the treatment effect directly, but to learn a common description of the units that can support both factual and counterfactual outcome prediction.

The representation layers are shared across treatment groups, allowing the model to exploit common predictive structure. The final outcome layers are separated, allowing the response to treatment to differ between groups. This architecture occupies an intermediate position between two independent outcome models and a single network that receives treatment as an ordinary input.

4.2 TARNet

The *Treatment-Agnostic Representation Network*, or TARNet (Shalit et al. 2017), consists of a shared representation network followed by the two heads in Equation (28). Its parameters are learned by minimizing the factual prediction loss together with an ordinary model-complexity penalty:

$$\mathcal{L}_{\text{TARNet}}(h, \Phi) = \hat{\epsilon}_F(h, \Phi) + \lambda R(h, \Phi). \quad (30)$$

The first term represent the factual loss from the respectively head, while the second is a complexity term. TARNet improves on two completely separate response models by allowing features to be learned jointly in the same space.

Nevertheless, the factual loss does not control the geometry of the treated and control

distributions in the latent space. Consider the following treatment-specific representation distributions:

$$p_{\Phi}^1(r) = p(\Phi(X) = r \mid T = 1) \quad \text{and} \quad p_{\Phi}^0(r) = p(\Phi(X) = r \mid T = 0). \quad (31)$$

If they remain very different, each head may still be evaluated on representations that are poorly supported by the observations used to train it. TARNet shares predictive information, but it does not explicitly reduce counterfactual distribution shift.

4.3 PEHE and the Need for Balanced Representations

TARNet is trained to predict the outcomes observed under the treatments actually received. The target of HTE estimation, however, is not the factual outcome itself, but the difference between the two treatment-specific predictions. A model may therefore achieve a small factual prediction error while still estimating treatment effects poorly, especially when one of the two outcome heads must extrapolate to regions that are weakly represented in its training data.

A common measure of such an error is the *Precision in Estimation of Heterogeneous Effect* (PEHE) (Hill 2011):

$$\epsilon_{\text{PEHE}}(h, \Phi) = \mathbb{E}_X \left[(\hat{\tau}(X) - \tau(X))^2 \right]. \quad (32)$$

PEHE measures the mean squared difference between the estimated and true treatment-effect functions over the target covariate distribution. A small value indicates that the model captures not only the ATE, but also how the effect varies across covariate profiles.

This distinction is important because errors in HTEs can cancel when only a population average is considered. For example, a model may overestimate the treatment effect for one subgroup and underestimate it for another, producing an accurate ATE while failing to recover the underlying heterogeneity. PEHE penalizes these subgroup-specific errors rather than considering only their average.

A principal limitation of PEHE is that its computation requires knowledge of the true treatment-effect function $\tau(x)$. Because only one potential outcome is observed for each unit in real-world data, PEHE cannot be evaluated directly. It is mainly used in simulations and semi-synthetic benchmarks, where the outcome-generating process provides both potential outcomes or the true conditional effect by construction. For the same reason, PEHE cannot normally be used as a direct training objective or as a standard validation criterion in real applications.

This creates a central theoretical problem. The quantity that should ideally be minimized is unavailable, whereas TARNet is trained only through errors on observed outcomes. The analysis of Shalit et al. (2017) connects these two levels by deriving an upper bound on PEHE. The bound shows that treatment-effect estimation error can be controlled through two components: accurate factual outcome prediction and similarity between the treated and control distributions in the learned representation space.

This result explains the transition from TARNet to CFR. In the same paper, Shalit et al. (2017) make this transition by introducing CFRNet. TARNet optimizes only the prediction component. CFR adds an explicit penalty for the discrepancy between the two latent treatment distributions, with the aim of reducing the extrapolation required when each outcome head is used to predict counterfactual outcomes.

This is done by introducing an Integral Probability Metric in the loss function.

4.4 Integral Probability Metrics

An Integral Probability Metric (IPM) measures the discrepancy between two distributions through the largest difference in expectation over a chosen class of test functions.

Definition 4.1 (Integral Probability Metric). Let P and Q be probability distributions on a space $\mathcal{S}$, and let $\mathcal{G}$ be a family of real-valued functions on $\mathcal{S}$. Their IPM is:

$$\text{IPM}_{\mathcal{G}}(P, Q) = \sup_{g \in \mathcal{G}} |\mathbb{E}_{Z \sim P}[g(Z)] - \mathbb{E}_{Z \sim Q}[g(Z)]|. \quad (33)$$

The function class $\mathcal{G}$ determines which differences the metric can detect. A small class may compare only a few moments, while a sufficiently rich class can distinguish any two different distributions. The choice of $\mathcal{G}$ also determines whether the resulting discrepancy can be estimated and differentiated efficiently during neural-network training.

In causal representation learning, the two distributions are p_{Φ}^1 and p_{Φ}^0 . If the expected outcome loss, expressed as a function of the representation, belongs to a scaled version of $\mathcal{G}$, then the IPM bounds how much its expectation can change when moving from the treated latent distribution to the control latent distribution. This is precisely the change of distribution required to relate factual and counterfactual risks.

Two choices of function class are particularly important. The unit ball of a reproducing kernel Hilbert space yields the Maximum Mean Discrepancy, while the class of one-Lipschitz functions yields the first Wasserstein distance. These metrics formalize two different notions of distributional similarity.

4.5 Maximum Mean Discrepancy

Let $k : \mathcal{R} \times \mathcal{R} \rightarrow \mathbb{R}$ be a positive-semidefinite kernel with associated reproducing kernel Hilbert space (RKHS) $\mathcal{H}_k$. The kernel corresponds to an implicit feature map φ satisfying:

$$k(r, r') = \langle \varphi(r), \varphi(r') \rangle_{\mathcal{H}_k}. \quad (34)$$

A probability distribution P can be represented by its mean embedding, computed as:

$$\mu_P = \mathbb{E}_{R \sim P}[\varphi(R)] \in \mathcal{H}_k. \quad (35)$$

When $\mathcal{G}$ is the unit ball of $\mathcal{H}_k$, the corresponding IPM is the Maximum Mean Discrepancy (Gretton et al. 2012):

$$\text{MMD}_k(P, Q) = \|\mu_P - \mu_Q\|_{\mathcal{H}_k}. \quad (36)$$

Thus, MMD measures the distance between the mean embeddings of the two distributions in the kernel feature space. When the kernel is characteristic, the embeddings are equal if and only if the underlying distributions are equal. Here, *mean* refers to an expectation in the kernel feature space, not merely to the coordinate-wise mean in the original space: the feature map may encode infinitely many nonlinear features.

The choice of kernel is substantive. A linear kernel reduces MMD to the Euclidean distance between latent means, whereas characteristic kernels such as the Gaussian kernel can detect broader distributional differences. Shalit et al. (2017) derive their framework for the RKHS-unit-ball IPM and report a squared linear-MMD variant in their experiments.

Richer kernels may provide stronger discrimination, but their behaviour depends on bandwidth, sample size, latent dimension, and minibatch estimation noise.

4.6 Wasserstein Distance

The first Wasserstein distance compares two distributions through the minimum cost of transporting the probability mass of one distribution into the other (Villani 2009). To define this cost, consider every possible way of matching mass from P with mass from Q while preserving both original distributions. Each such matching plan is called a *coupling*, and the set of all possible couplings is denoted by $\Pi(P, Q)$. The distance is then defined as:

$$W_1(P, Q) = \inf_{\gamma \in \Pi(P, Q)} \mathbb{E}_{(R, S) \sim \gamma} [d(R, S)]. \quad (37)$$

Here, d is the ground distance in the representation space, and γ specifies a particular coupling. The Wasserstein distance selects the coupling with the lowest expected transport cost.

This interpretation makes Wasserstein distance sensitive to the geometry of the latent space. Moving probability mass between nearby representations has a lower cost than moving it between distant ones. Therefore, two distributions may be considered relatively close even when their samples do not overlap exactly, provided that their probability mass is concentrated in nearby regions.

Kantorovich–Rubinstein duality provides an equivalent perspective: instead of minimizing over transport plans, one searches for the one-Lipschitz function that maximally distinguishes the two distributions. Consequently, the first Wasserstein distance can also be expressed as an Integral Probability Metric over the class of one-Lipschitz functions (Santambrogio 2015):

$$W_1(P, Q) = \sup_{\|g\|_{\text{Lip}} \leq 1} |\mathbb{E}_P[g(R)] - \mathbb{E}_Q[g(R)]|. \quad (38)$$

It can therefore be used in CFR to measure the discrepancy between the treated and control representation distributions.

MMD and Wasserstein capture distributional discrepancy in different ways. MMD compares distributions through differences in kernel mean embeddings, whereas Wasserstein measures the geometric cost of transporting one distribution into the other. The choice between them therefore introduces different assumptions about which properties of the learned representation space should determine balance.

Consequently, Wasserstein distance is the IPM generated by one-Lipschitz test functions. Unlike MMD, it explicitly depends on a geometry through the ground distance. Moving mass between nearby representations is cheaper than moving it between distant ones.

4.7 A Generalization Bound for Treatment-Effect Estimation

IPMs play a central role in the main theoretical result of Shalit et al. (2017), which formalizes the role of distributional balance. Their analysis assumes strong ignorability and considers a twice-differentiable Φ . The invertibility assumption prevents the theoretical transformation from discarding information and permits the treated and control densities to be related through a change of variables. It is an analytical assumption, not a property guaranteed by ordinary neural-network implementations.

Before stating the bound, it is useful to distinguish the factual prediction errors of the two treatment groups. Under squared loss, the expected factual error among treated units is:

$$\epsilon_F^{t=1}(h, \Phi) = \mathbb{E} \left[(Y(1) - h_1(\Phi(X)))^2 \mid T = 1 \right], \quad (39)$$

The corresponding error among control units is:

$$\epsilon_F^{t=0}(h, \Phi) = \mathbb{E} \left[(Y(0) - h_0(\Phi(X)))^2 \mid T = 0 \right]. \quad (40)$$

These terms measure how accurately the two outcome heads predict outcomes in the populations from which their training observations are actually obtained. By consistency, $Y(1) = Y$ for treated units and $Y(0) = Y$ for control units, so their empirical counterparts can be estimated from the observed data.

The bound also depends on the distributions of treated and control units after applying the representation Φ (p_Φ^1 and p_Φ^0), and on their discrepancy measured through an IPM.

Under squared loss and regularity conditions ensuring that the represented expected-loss functions belong to a scaled function class $\mathcal{G}$, Theorem 1 of Shalit et al. (2017) express the bound:

$$\begin{aligned} \epsilon_{\text{PEHE}}(h, \Phi) &\leq 2 \left(\epsilon_{CF}(h, \Phi) + \epsilon_F(h, \Phi) - 2\sigma_Y^2 \right) \\ &\leq 2 \left(\epsilon_F^{t=0}(h, \Phi) + \epsilon_F^{t=1}(h, \Phi) + B_\Phi \text{IPM}_\mathcal{G} \left(p_\Phi^1, p_\Phi^0 \right) - 2\sigma_Y^2 \right). \end{aligned} \quad (41)$$

Here $B_\Phi > 0$ depends on the relationship between the expected loss, the representation, and the selected function class, while σ_Y^2 is a lower-bound variance term.

The value of Equation (41) includes three main contributions:

1. Factual outcome error in the control population.
2. Factual outcome error in the treated population.
3. Discrepancy between the two treatment-group distributions in the learned representation space.

The bound explains why minimizing factual prediction error alone may be insufficient. A model can fit observed outcomes well while producing unreliable predictions after crossing from one treatment distribution to the other. It also explains why balancing alone is insufficient: a representation with small IPM but poor outcome prediction still has a loose bound.

Several qualifications are essential. First, an upper bound is not an equality and does not imply that every decrease in empirical IPM produces a corresponding decrease in true PEHE. Second, the coefficient B_Φ is generally unavailable during training. Finally, the result presupposes the causal identification assumptions; it does not protect against unmeasured confounding or structural lack of overlap.

4.8 Counterfactual Regression

The generalization bound motivates an end-to-end objective that combines factual prediction and latent distribution balance. Let u be the treated proportion:

$$u = \frac{1}{n} \sum_{i=1}^n T_i. \quad (42)$$

Let's also define treatment-group weights as:

$$w_i = \frac{T_i}{2u} + \frac{1 - T_i}{2(1 - u)}. \quad (43)$$

These weights give equal aggregate importance to treated and control factual losses even when the group sizes differ. CFR propose to minimize:

$$\begin{aligned} \mathcal{L}_{\text{CFR}}(h, \Phi) = & \frac{1}{n} \sum_{i=1}^n w_i \ell(Y_i, h_{T_i}(\Phi(X_i))) + \lambda R(h, \Phi) \\ & + \alpha \text{IPM}_{\mathcal{G}}(\hat{p}_{\Phi}^1, \hat{p}_{\Phi}^0). \end{aligned} \quad (44)$$

The final term is estimated from the treated and control representations in the sample or minibatch. TARNet is the special case $\alpha = 0$; CFRNet uses $\alpha > 0$, with either an MMD- or Wasserstein-based penalty in the original experiments. An overview of the CFRNet architecture is visible in Figure 1.

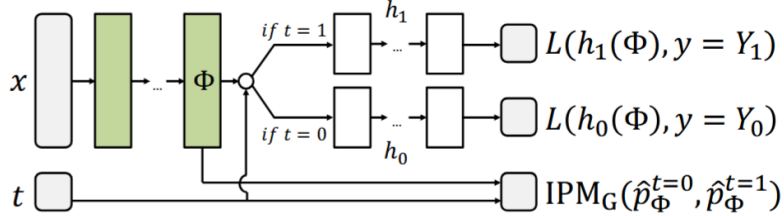


Figure 1: Overview of the CFRNet architecture. A shared network maps the covariates into a learned representation, which is passed to treatment-specific outcome heads, while an IPM penalty encourages balance between the treated and control representation distributions. Reproduced from Shalit et al. (2017).

The parameter α absorbs the unavailable scale factor B_{Φ} and controls a prediction/balance trade-off. When α is too small, the model behaves like ordinary factual outcome regression and may retain severe latent imbalance. When it is too large, the representation may sacrifice outcome-relevant distinctions merely to make treatment groups similar. In the extreme, mapping every observation to the same point would achieve zero empirical discrepancy but make accurate HTE prediction impossible.

The original CFR objective therefore constrains the scale of Φ , using projection or fixed-scale batch normalization, to prevent arbitrary rescaling from reducing the IPM term without meaningfully changing the representation. More generally, the need for scale control illustrates that balance penalties depend on the geometry chosen by the model.

The method is trained jointly by stochastic gradient descent. The factual loss updates the shared representation and the observed-treatment head, while the IPM term updates the representation using both groups. At inference time, both heads are evaluated for each covariate vector and their difference yields $\hat{\tau}(x)$.

4.9 Evaluation and Scope

Shalit et al. (2017) evaluate TARNet and CFR on the semi-synthetic Infant Health and Development Program benchmark and on the Jobs dataset. Their results support the practical relevance of the balance penalty, particularly when treatment assignment produces substantial covariate imbalance. At the same time, they illustrate a general difficulty in causal machine learning: hyperparameter selection for PEHE is non-trivial because true individual or conditional effects are unavailable in observational validation data.

The framework contributes an important connection between causal estimation and domain adaptation, but its scope should remain explicit. CFR addresses finite-sample counterfactual generalization under observed-covariate imbalance. It does not identify effects under hidden confounding, create support where one treatment is structurally absent, or guarantee that a balanced representation preserves every effect modifier. Moreover, the theoretical analysis is stated for invertible representations, whereas the practical neural networks are optimized through a surrogate objective and may compress information.

These limitations lead to the next question. Global alignment of treated and control distributions can reduce covariate shift, but it does not determine which local relationships the latent space should preserve. Later methods therefore explore self-supervised balancing, propensity-related objectives, and contrastive structures intended to make the representation more informative for treatment-effect heterogeneity rather than merely treatment-invariant.

5 Beyond Global Covariate Balance

The CFR objective in Equation (44) encourages the treated and control distributions to become similar after representation learning. This global alignment is useful for reducing counterfactual distribution shift, but it does not fully determine the geometry of the latent space. Two representations may have a comparable IPM while preserving very different local relationships. In particular, aggressive treatment-invariance may remove covariates that predict the outcome or modify the treatment effect. The objective is therefore not balance alone, but a representation that balances the groups while retaining the distinctions needed to estimate heterogeneous responses.

Recent methods explore alternative signals for constructing such representations. BCAUSS targets covariate balance directly through a self-supervised objective, whereas HERMES introduces pairwise information about estimated treatment-response similarity. These methods illustrate a broader transition from matching entire treatment-group distributions to asking which relationships among individuals should be preserved.

5.1 Self-Supervised Auto-Balancing with BCAUSS

BCAUSS, short for *Balancing Covariates Automatically Using Self-Supervision* (Tesei et al. 2023), uses the three-task architecture also adopted by earlier Dragonnet (Shi et al. 2019): a shared representation feeds two treatment-specific outcome heads and an additional scalar head $g_\theta(X)$, visible in Figure 2.

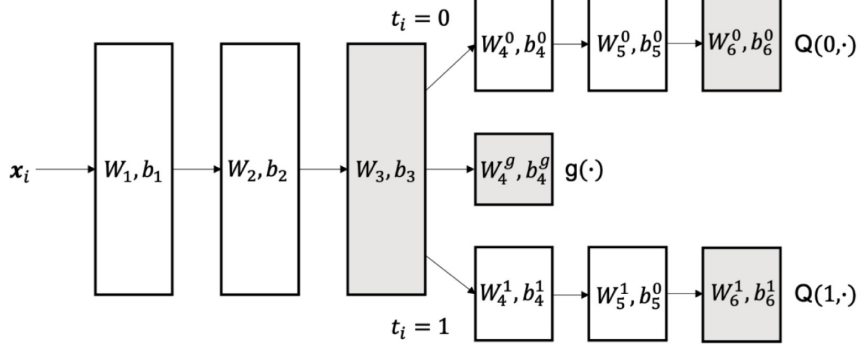


Figure 2: Overview of the BCAUSS architecture, in which a shared representation feeds two treatment-specific outcome heads and a self-supervised balancing head. Reproduced from Tesei et al. (2023).

The outcome heads are trained through factual regression. The third head is not trained to predict treatment assignment; instead, it serves as main tool for per-unit alignment. While Dragonnet trains it to predict the PS used in the balancing, BCAUSS optimizes it to make inverse-weighted treated and control covariate means similar.

For covariate k , let $x_{i,k}$ denote its value for unit i , BCAUSS defines the balancing objective:

$$\mathcal{L}_{\text{BAL}}(\theta) = \frac{1}{d} \sum_{k=1}^d \left(\frac{\sum_{i=1}^n T_i x_{i,k} / g_\theta(X_i)}{\sum_{i=1}^n T_i / g_\theta(X_i)} - \frac{\sum_{i=1}^n (1 - T_i) x_{i,k} / \{1 - g_\theta(X_i)\}}{\sum_{i=1}^n (1 - T_i) / \{1 - g_\theta(X_i)\}} \right)^2 \quad (45)$$

The term is *self-supervised* because it uses the observed covariates and treatment assignments but no unobserved potential outcomes. It is *auto-balancing* because the weighting function is learned by directly minimizing residual imbalance rather than only by maximizing treatment-classification accuracy. The complete training criterion combines Equation (45) with the factual outcome loss, so the shared representation is simultaneously encouraged to predict outcomes and support balanced comparisons.

This distinction becomes relevant when overlap is poor. A treatment classifier may accurately reproduce a nearly deterministic assignment mechanism while producing extreme scores and unstable weights. BCAUSS instead minimizes observed covariate imbalance directly, which can reduce bias when both treatment groups are present but unevenly represented. In the reported experiments, BCAUSS achieved lower bias than the considered baselines, particularly in settings with weak positivity. However, it cannot resolve structural violations of positivity: if one treatment is never observed, no balancing method can recover the missing counterfactual information. BCAUSS therefore mitigates the statistical consequences of limited empirical overlap rather than identifying effects in unsupported regions.

5.2 Effect-Aware Contrastive Representations

Global distribution matching and covariate balancing still define similarity mainly through the observed covariates. *Heterogeneous Effects Representation with Matched Embeddings using Siamese Networks* (HERMES) introduces a different signal: similarity in estimated treatment effects (Zaccagnino et al. 2026). The method uses a Siamese architecture and dynamically constructs pairs of individuals whose preliminary effect estimates are similar or dissimilar. The architecture overview is visible in Figure 3. A contrastive objective then encourages nearby embeddings to correspond to similar treatment responses rather than merely similar covariate profiles.

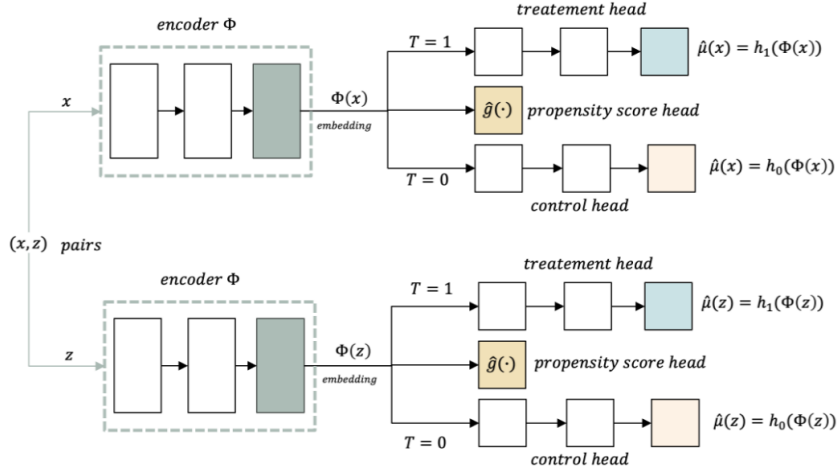


Figure 3: Overview of the HERMES architecture, which uses Siamese encoders and a contrastive objective to organize representations according to estimated treatment-effect similarity. Reproduced from Zaccagnino et al. (2026).

This shifts the objective of representation learning. Rather than focusing solely on global alignment between the treated and control populations, HERMES seeks to construct a local geometry that reflects treatment-effect heterogeneity. The contrastive component is used during training, while treatment-specific outcome predictions retain the usual inference-time form. Reported experiments on IHDP and Jobs indicate improved HTE estimation without additional inference cost.

The approach also introduces a new dependency: pair construction relies on estimated rather than observed unit-level effects. Errors in those preliminary estimates may therefore influence the geometry learned by the Siamese network. More generally, effect-aware pairing does not remove the identification assumptions established in Section 2; it provides an inductive bias for organizing the representation once those assumptions are adopted.

5.3 From Balance to Representation Geometry

As summarized in Table 1, the evolution is not simply toward stronger distribution matching. It is toward richer criteria for deciding what the representation should preserve. This observation motivates the next question: whether external semantic information can provide an additional, domain-informed signal while remaining subordinate to explicit causal assumptions and statistical validation.

Table 1: Main signals used to shape causal representations.

Method	Representation signal	Main idea
TARNet	Factual outcome loss	Share predictive structure across treatment groups.
CFR	Global IPM penalty	Reduce discrepancy between treated and control latent distributions.
BCAUSS	Weighted covariate balance	Learn a balancing score by directly minimizing empirical imbalance.
HERMES	Pairwise estimated-effect similarity	Organize local neighborhoods around treatment-response similarity.

6 Semantic Information and Large Language Models

The methods considered so far learn exclusively from the observed covariates, treatment assignments, and factual outcomes. In many scientific applications, however, the variables also have textual descriptions, domain-specific names, or associated documents. Large language models (LLMs) can transform this material into numerical representations that reflect regularities learned during pre-training. The relevant question is not whether an LLM can replace a causal estimator, but whether such semantic information can provide a useful inductive bias while remaining subordinate to explicit identification assumptions and statistical validation.

6.1 Direct Causal Outputs and Their Limits

Current research examines the application of LLMs across several core tasks in causal inference, including causal-relation discovery, counterfactual generation, and treatment-effect estimation. Some literature, surveyed by Liu et al. (2025), suggests that the world knowledge embedded within LLMs could support traditional causal methods through data augmentation to mitigate sample imbalance or by supporting the assumption of unconfoundedness.

However, systematic empirical evaluations of the deductive reasoning capabilities of LLMs reveal critical fallacies and biases (Nitish Joshi and He 2024):

- **Position Heuristic:** LLMs frequently infer a causal relation ($X \rightarrow Y$) based merely on the sequential order of entity mentions in text (i.e., assuming X mentioned before Y implies X causes Y), ignoring the true semantics of the relation.
- **Post Hoc Fallacy:** When mention order is randomized to reduce position bias, models may instead exhibit the post hoc fallacy, incorrectly inferring causation from temporal precedence (i.e., assuming that “ X precedes Y ” implies “ X causes Y ”).
- **Spatial, Temporal, and Counterfactual Reasoning:** While LLMs can correctly deduce the absence of causal relations using temporal and spatial constraints, they struggle to infer the presence of causal relations from counterfactual statements.
- **Impact of Model Scaling:** Increasing model size does not resolve reliance on the position heuristic and actually worsens performance regarding the post hoc fallacy (inverse scaling).

Experimental results suggest that LLMs may struggle to infer novel causal knowledge autonomously, relying predominantly on memorized causal facts extracted during pretraining.

6.2 Semantic Embeddings as Side Information

A more limited role is to use the LLM as a representation provider. Let s_j denote a textual description associated with feature j , and let E_{LLM} be a fixed encoder. The semantic representation is then:

$$z_j^{\text{sem}} = E_{\text{LLM}}(s_j). \quad (46)$$

These vectors may encode similarities that are absent from a table of numerical measurements. For example, two clinical variables may describe related physiological processes despite having different scales or little marginal correlation. A simple semantic kernel (Basili et al. 2005; Altinel et al. 2015) can be defined as:

$$k_{\text{sem}}(s_j, s_\ell) = \langle z_j^{\text{sem}}, z_\ell^{\text{sem}} \rangle. \quad (47)$$

For a fixed encoder, this inner product is a positive-semidefinite kernel and supplies a learned notion of feature similarity. It does not, however, imply that the features are causally related or interchangeable. Semantic similarity may reflect shared terminology, research attention, or common context rather than confounding, mediation, or effect modification.

In causal representation learning, such information could be used in several ways. It might regularize which features are grouped in the latent space, guide pair construction in a contrastive objective, or define a prior over directions expected to contain relevant signal. These possibilities are best regarded as research hypotheses. Their value depends on whether the semantic structure is aligned with the target outcome and treatment mechanism, and that alignment must be evaluated empirically rather than assumed.

6.3 Priors under Statistical Control

An instructive example is provided by Bryan et al. (2025), who use embeddings derived from textual gene descriptions in genomics hypothesis testing. The principal subspace of the embedding matrix is treated as prior information about directions in which biological signals may concentrate. Within frequentist and Bayesian frameworks, the resulting procedures can concentrate statistical power along this subspace while maintaining type-I error control, even when the LLM-derived information is uninformative. Their empirical examples report higher power than classical counterparts when the biological signal is aligned with the embedding subspace.

The work, however, concerns hypothesis testing rather than treatment-effect estimation. Its relevance here is therefore methodological rather than evidential: semantic information is introduced indirectly, through a prior that changes efficiency, while the validity condition remains part of the statistical procedure. A similar principle could motivate causal estimators whose representation or regularization is informed by semantic structure but whose identification assumptions remain unchanged.

Several safeguards would be necessary. Semantic priors should be compared with uninformative or randomly constructed alternatives, and sensitivity to model, prompt, and embedding dimension should be reported. Sample splitting or cross-fitting may be needed to avoid leakage and overfitting. Finally, robustness to prior misspecification is essential: misleading semantic information should reduce efficiency rather than silently invalidate the causal analysis. Under these constraints, LLM-derived representations are most naturally viewed as auxiliary information, not as a source of counterfactual truth.

7 Discussion

The methods reviewed in this work address different layers of the treatment-effect estimation problem. The potential-outcomes framework defines the estimand and the assumptions under which it is identifiable, while outcome and propensity models estimate functionals of the observed-data distribution. Representation learning changes the statistical geometry in which those functions are learned.

7.1 Identification and Representation

Causal representations can reduce dimensionality, facilitate information sharing across treatment groups, and improve generalization when overlap is limited but non-zero. However, they cannot establish that no relevant confounder has been omitted or recover counterfactual responses in regions with no support. A small latent-space discrepancy is therefore a property of the fitted representation, not evidence that conditional exchangeability holds. The theory in Section 4 supports only a conditional interpretation: given the identification assumptions and regularity conditions, factual prediction error and representation imbalance jointly determine an upper bound on treatment-effect estimation error.

7.2 Balance and Information Preservation

The main design tension is between comparability by overlap and predictive sufficiency. If treated and control representations remain far apart, counterfactual predictions require extrapolation, but, at the same time, if the groups are forced to be indistinguishable too aggressively, the encoder may remove variables that predict the outcome or modify the treatment effect. TARNet, CFR, BCAUSS, and HERMES differ partly in the signal they use to negotiate this trade-off: factual prediction, global distributional distance, weighted covariate balance, or local estimated-effect similarity. None of these objectives uniquely appear to describe the ideal representation, because the relevant geometry depends on the estimand and data-generating process.

The transition from global to local criteria is nevertheless important. An aggregate IPM can be small even when neighborhoods that matter for heterogeneous response are poorly organized. Contrastive approaches attempt to preserve more targeted relationships, but they introduce dependence on pair construction and preliminary effect estimates. Semantic priors extend the same idea by using domain-derived similarity. Their advantage is the possibility of incorporating knowledge not visible in the numerical sample; their risk is that semantic proximity may be unrelated, or even opposed, to causal relevance.

7.3 Evaluation and Reliability

Model selection remains difficult because the true CATE is unavailable in real observational data. Semi-synthetic benchmarks provide known counterfactuals but inherit the assumptions of their outcome simulations. Factual prediction error is observable but can favour models that fit the received treatment while extrapolating poorly under the alternative. Balance diagnostics are useful but do not measure treatment-effect accuracy directly. Reliable evaluation should therefore combine multiple forms of evidence: benchmark performance, overlap and balance diagnostics, policy-oriented criteria where appropriate, uncertainty quantification, and sensitivity analyses for unmeasured confounding and modelling choices.

Concerning the sensitivity analysis, several attempts are found in literature to provide a meaningful assessment framework. Examples worth mentioning:

Rosenbaum Bounds: the Gamma Parameter (Γ) (Rosenbaum 2013) represents the maximum odds ratio that two matched individuals with the same observed covariates could differ in their odds of receiving the treatment due to an unobserved factor, or in other words it shows how strong an omitted variable’s effect must be to overturn your study’s conclusions. If $\Gamma = 1$, treatment assignment is unconfounded given the observed covariates. A value of 1.5 – 2.0 means high confounding instead.

E-Value: it calculates the minimum strength (minimum risk ratio) an unmeasured confounder must have with both the treatment and the outcome to nullify an observed association (VanderWeele and Ding 2017). A calculated E-value is interpreted by comparing it to known, measured confounders in that specific field of study.

Cinelli & Hazlett Framework: it is an extension of the omitted variable bias framework into a suite of sensitivity analysis tools by parameterizing the strength of unmeasured confounders using partial R^2 values, which describe the unique contribution of a specific independent variable to a regression model. It introduces the Robustness Value (RV), a practical metric that quantifies the minimum explanatory power unobserved confounders must share with both the treatment and the outcome to overturn a study’s estimated effects (Cinelli and Hazlett 2020).

As the methodological landscape expands and the causal inference framework continues to evolve, the same standards of empirical validation should be applied to information derived from LLMs. Improvements should be demonstrated against non-semantic baselines and across alternative encoders, prompts, and prior strengths. Ablations are particularly important because an embedding may help through generic dimension reduction rather than through domain knowledge. Procedures should also make clear which guarantees remain valid if the semantic prior is wrong.

Overall, the central problem is not merely to learn a balanced representation. It is to determine which distinctions should be removed, which information should be preserved, and how the causal relevance of the resulting geometry can be assessed when counterfactual ground truth is unavailable.

8 Conclusion

Estimating HTEs from observational data combines a causal identification problem with a difficult prediction problem. Potential outcomes make the target contrast explicit, while consistency, conditional exchangeability, and positivity determine whether it can be expressed through the observed distribution. Even when these assumptions are adopted, covariate imbalance makes each treatment-specific outcome model generalize beyond the population that supplied its factual training outcomes.

Causal representation learning addresses this second difficulty. TARNet shares predictive structure, CFR adds an explicit distributional penalty justified by an IPM-based generalization argument, and later methods improve the balancing signal through self-supervision or contrastive effect-aware geometry. These developments show that balance is not an end in itself: a useful representation must also preserve the information that determines outcomes in the overall distribution.

LLM-derived embeddings suggest a further source of inductive bias. They may encode semantic relationships that help define priors, regularizers, or local similarities, but they do not supply missing potential outcomes or validate causal assumptions. One of the most promising direction is therefore not to replace causal estimators with foundation models, but to incorporate semantic side information into procedures in which target estimand, assumptions, uncertainty, and failure modes remain explicit.

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